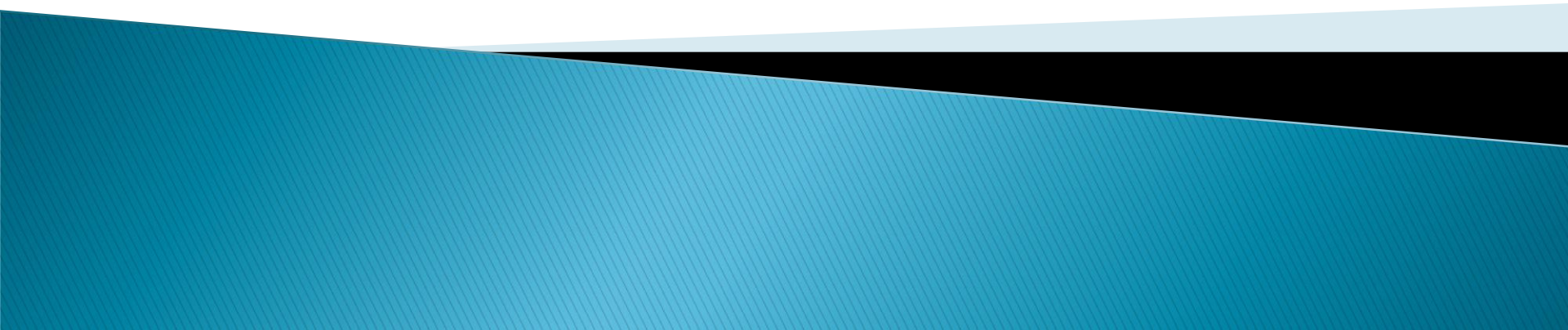


Computação Gráfica

AULA 12: Texturização

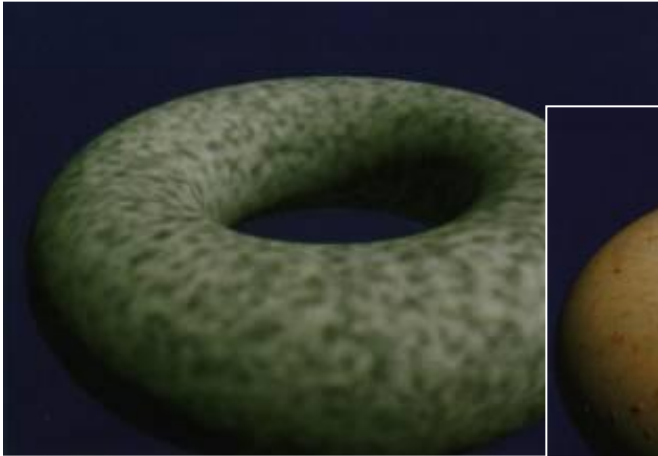
Prof. Paulo Bressan



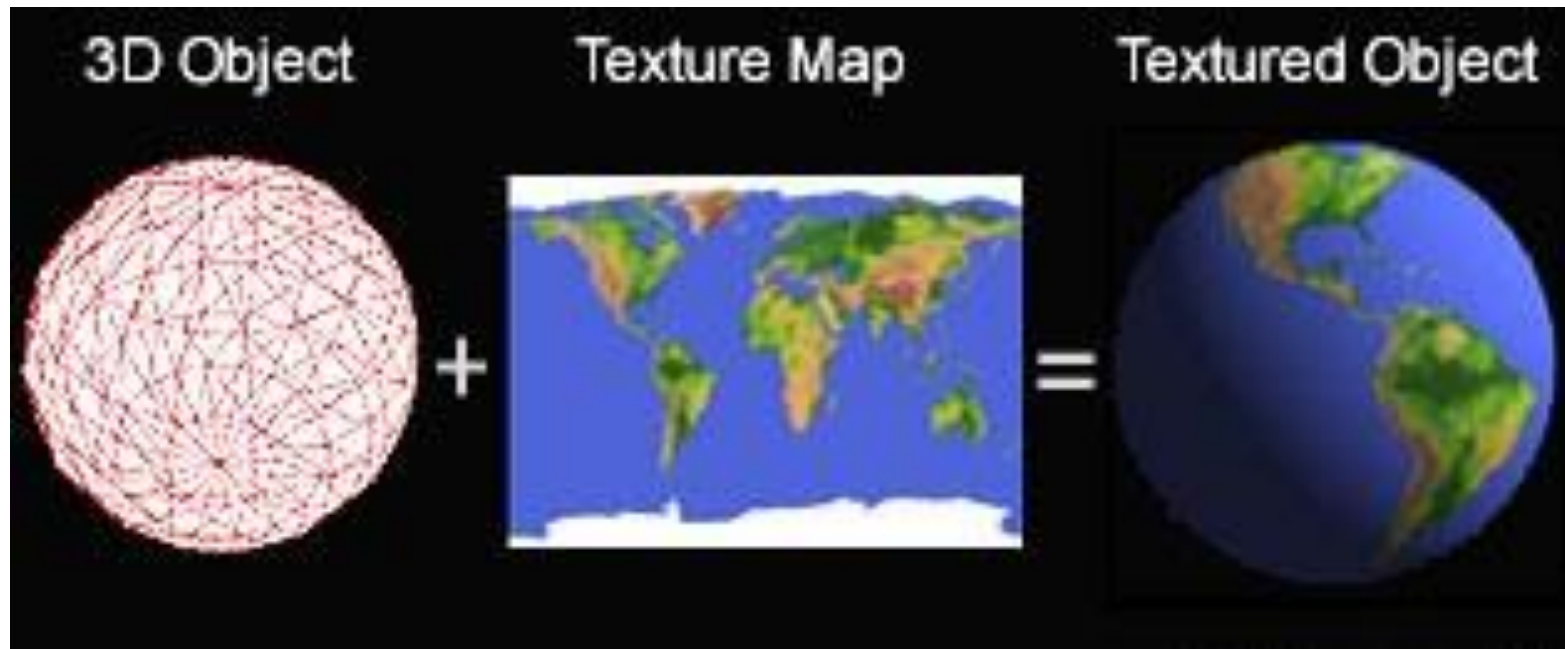
Exemplos



Exemplos



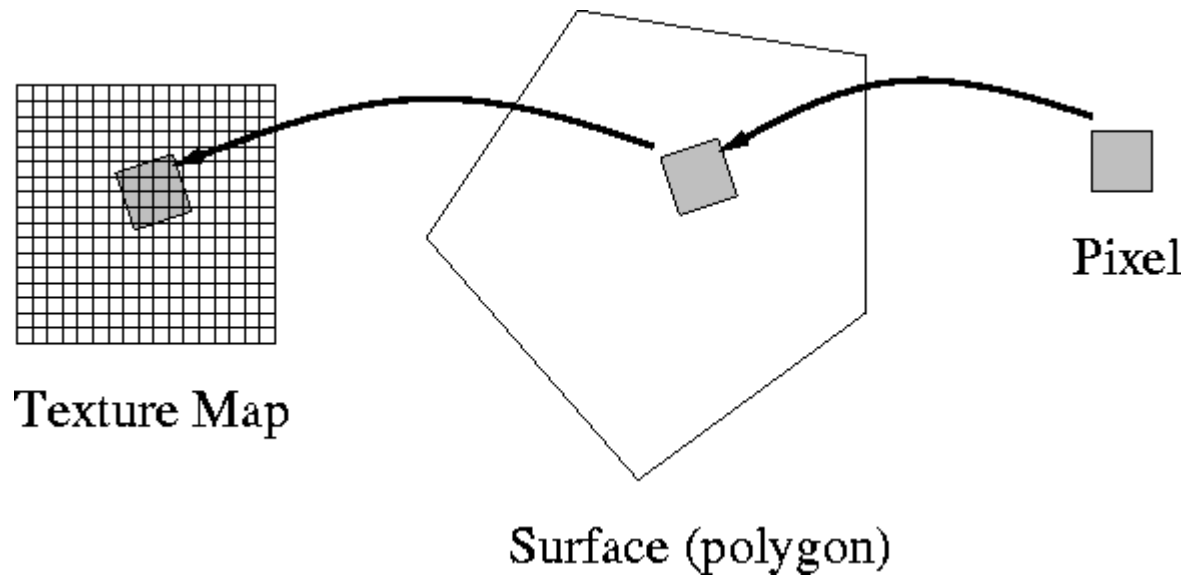
Exemplos



Texturização

- ▶ Permite aumentar o realismo do ambiente e das superfícies dos objetos
- ▶ Existem dois problemas principais:
 - Como gerar as texturas dos materiais?
 - Como texturizar o ambiente e os objetos?
 - Mapeamento de textura
 - Texturização sólida

Função de Mapeamento



Texture maps

Tom Porter's Bowling Pin



Reflection maps

Blinn and Newell, 1976



Environment maps



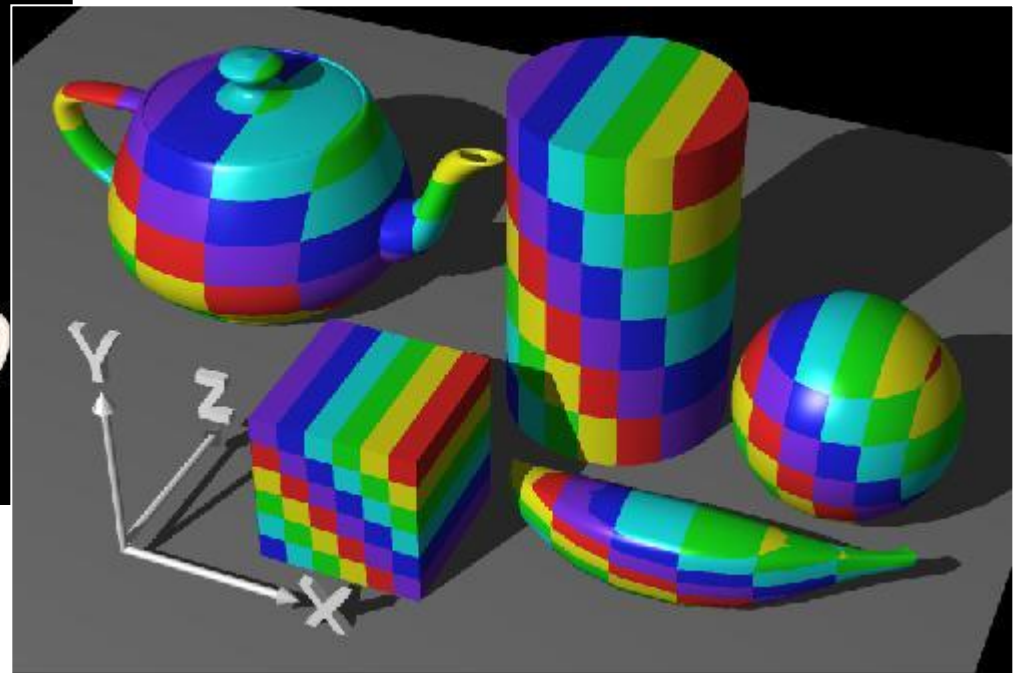
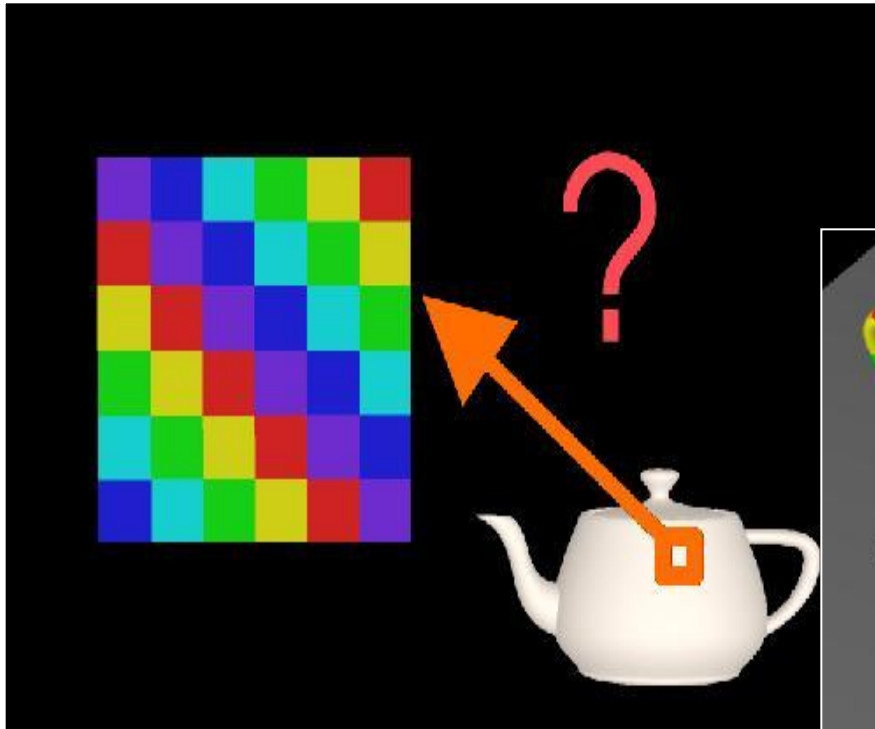
Ray Traced



Environment Map

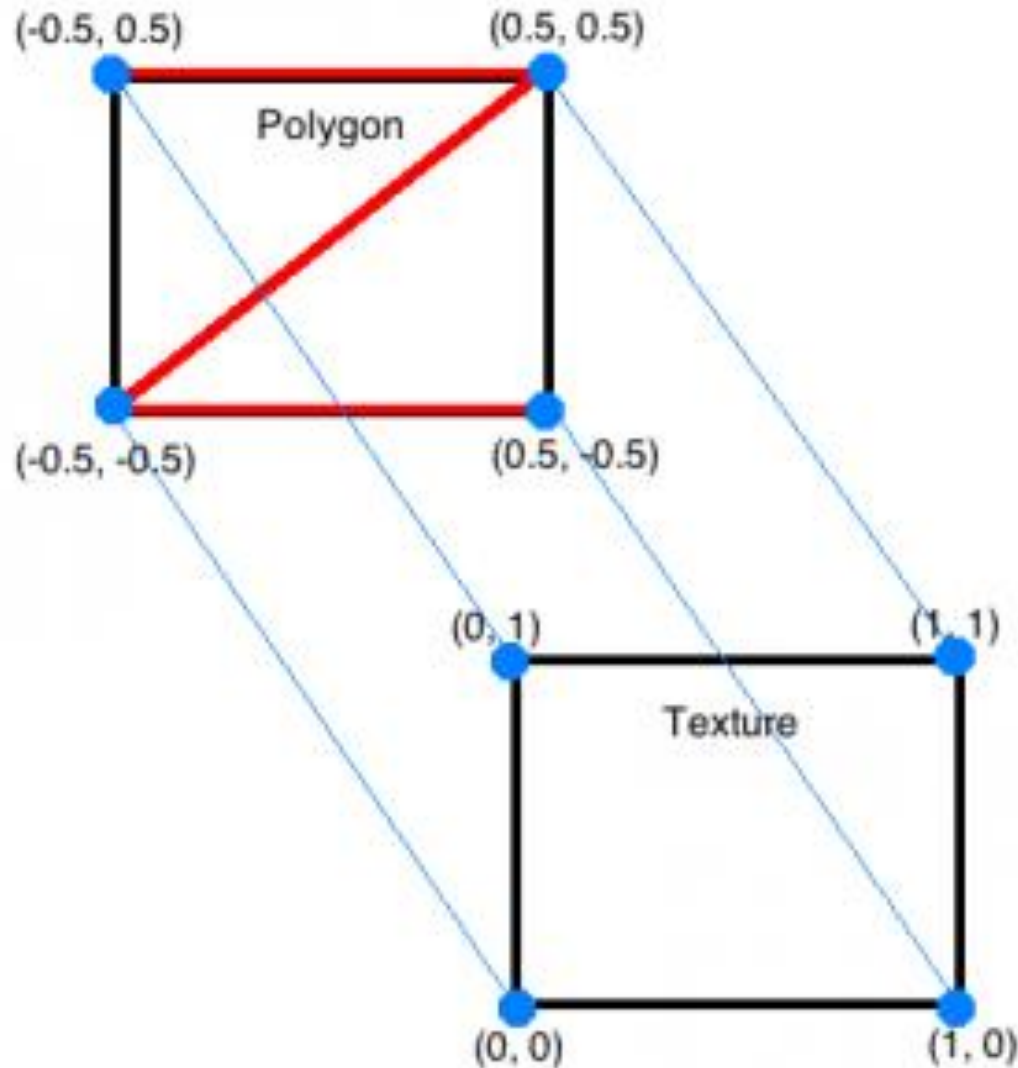
Mapeamento de Textura

Texturas Bidimensionais

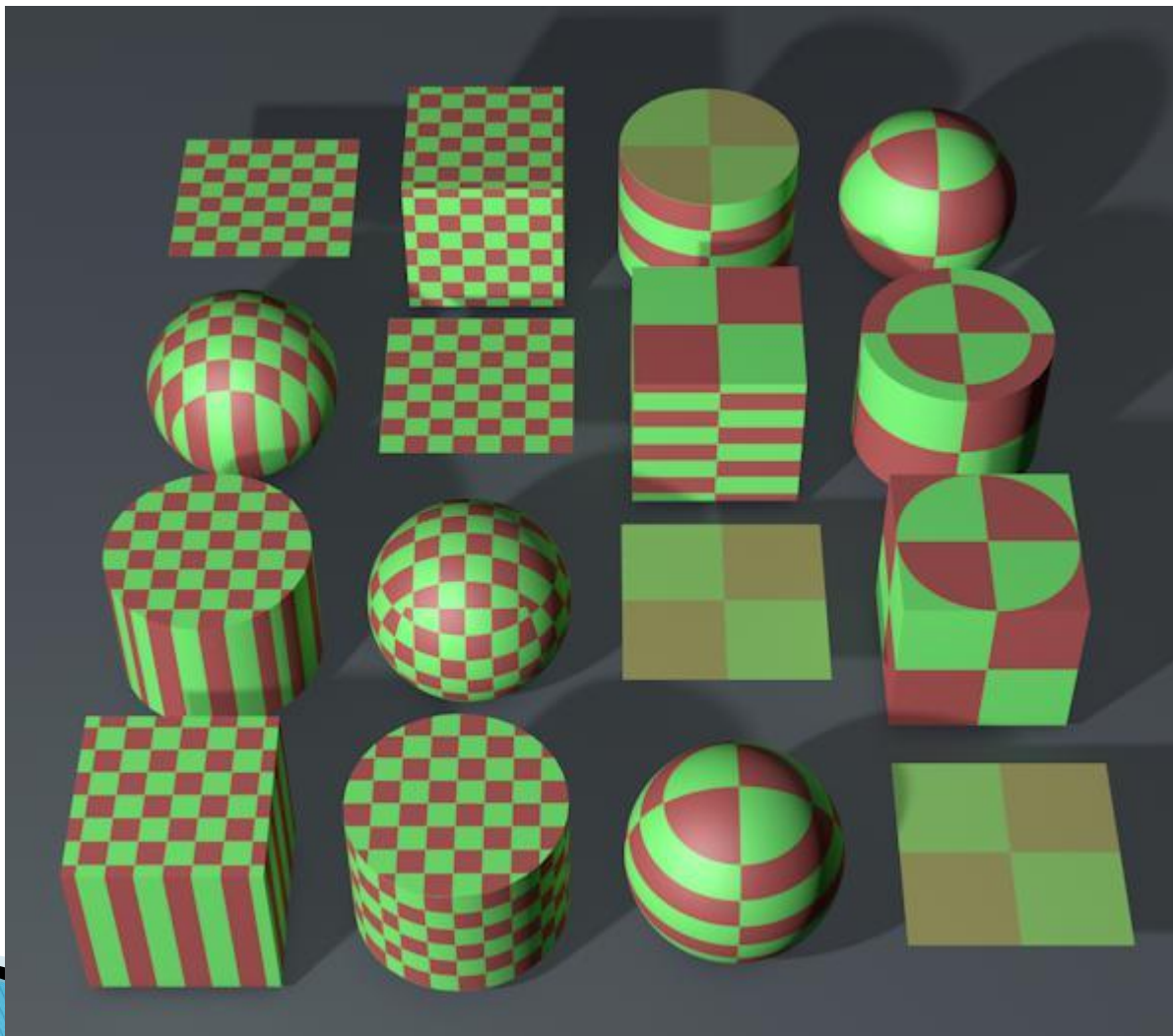


Mapeamento de Textura

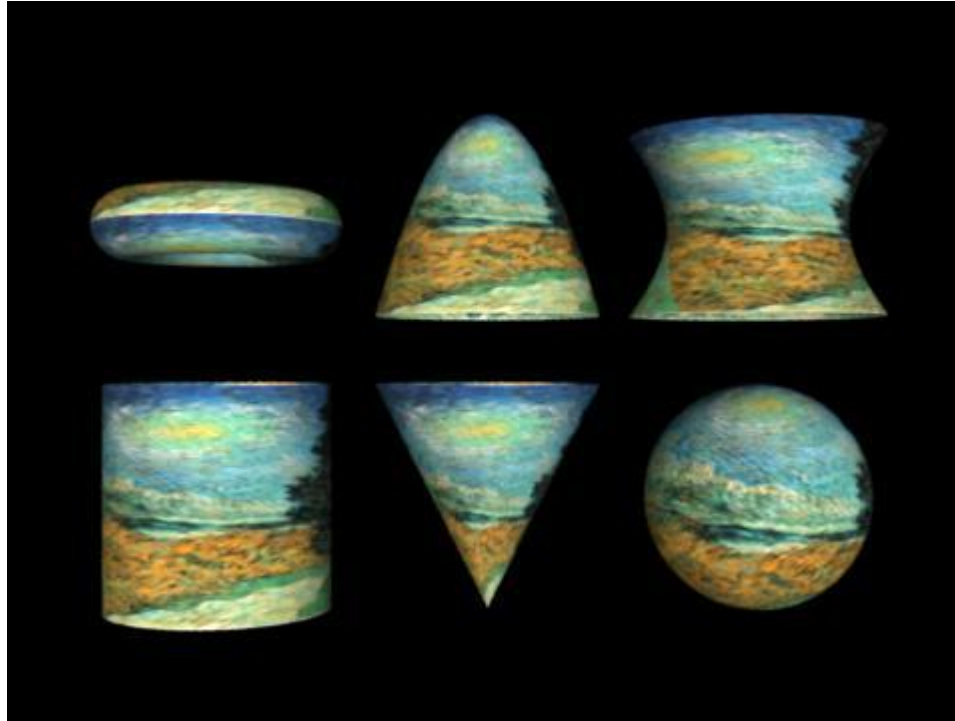
Texturas Bidimensionais



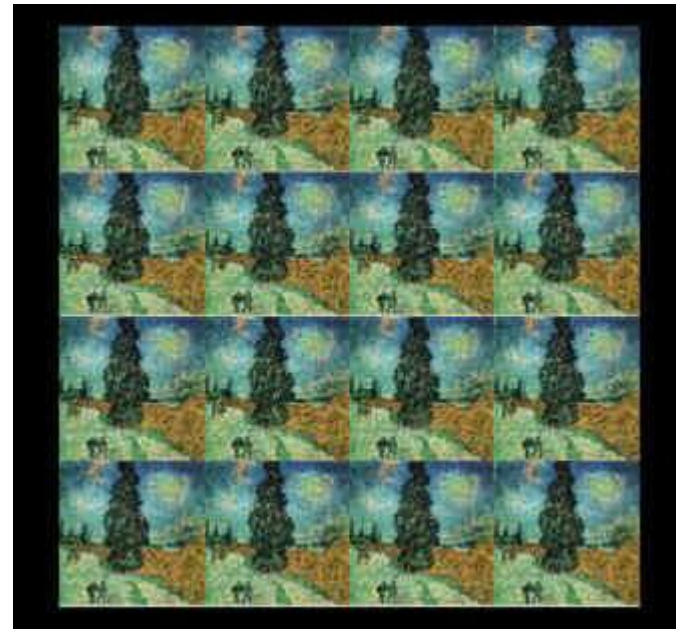
Mapeamento de Textura



Mapeamento de Textura

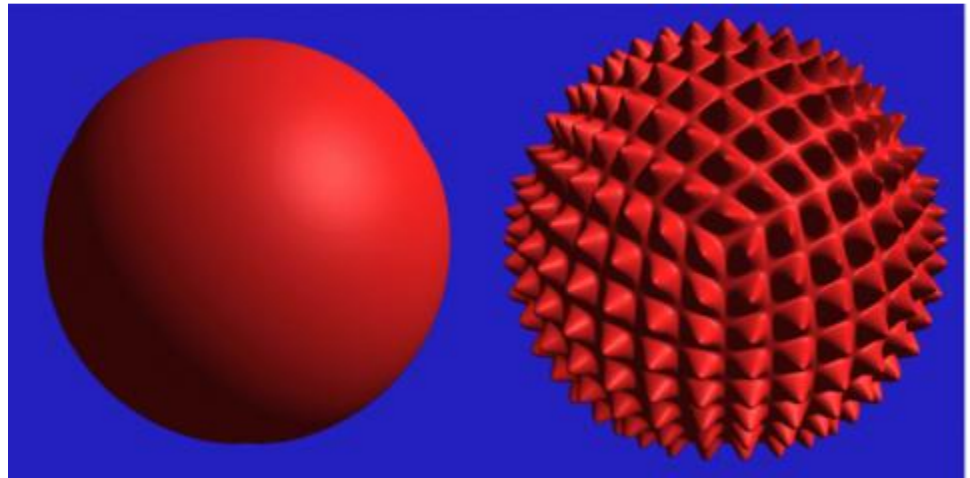
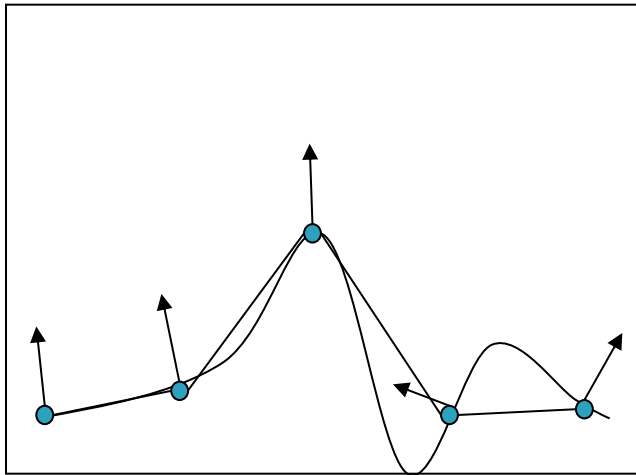


Mapeamento de Textura



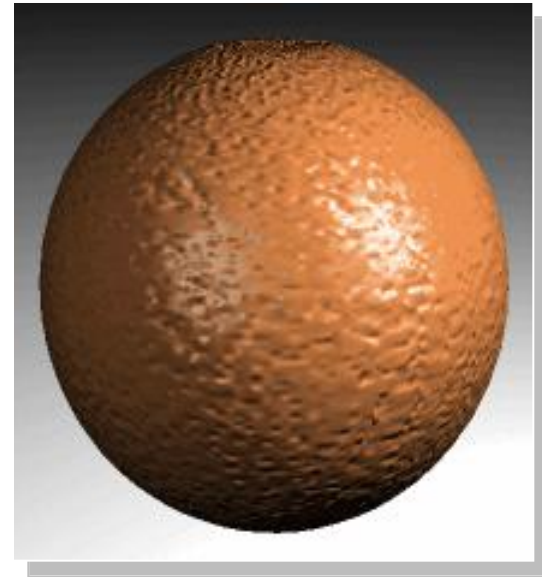
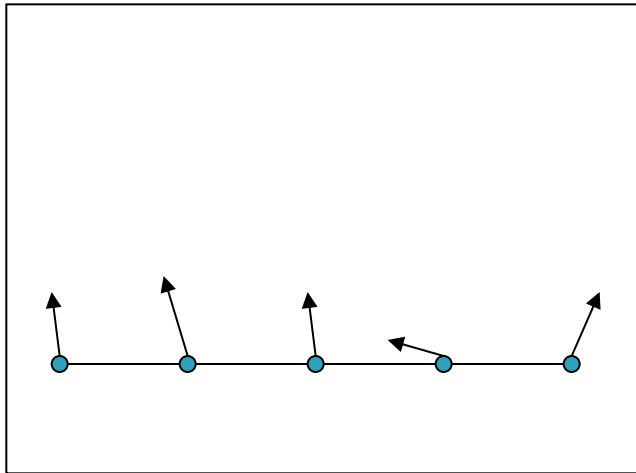
Displacement Mapping

Altera posição dos vértices da superfície do objeto. Requer subdivisão da malha para aumentar precisão.



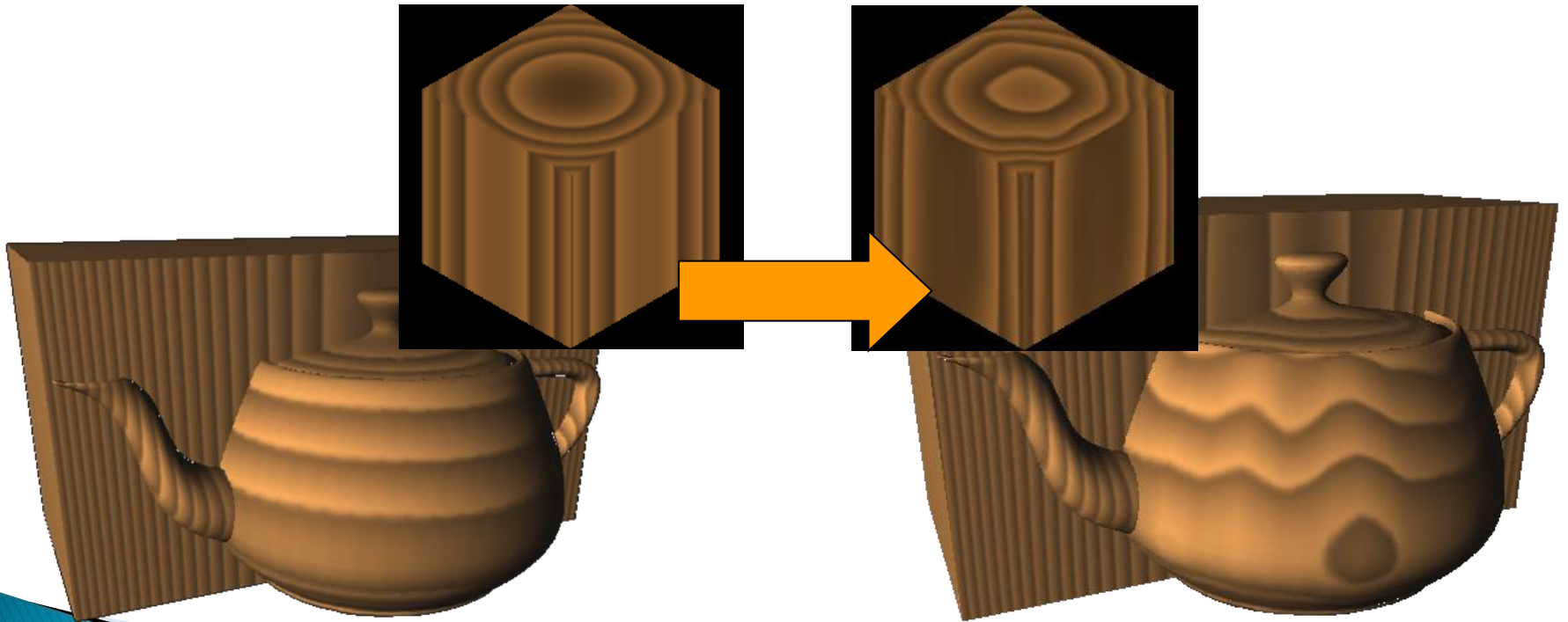
Bump Mapping

Altera somente a direção das normais.
Dá uma ilusão de displacement mapping.



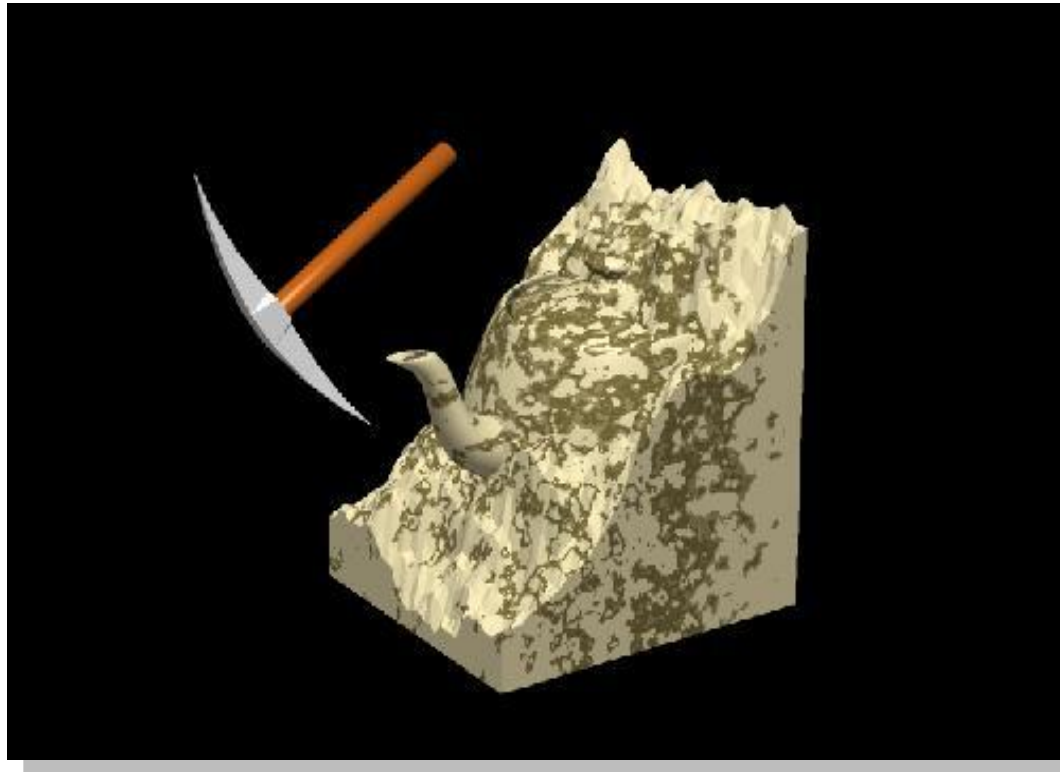
Composição de Ruídos

Parte-se de uma forma básica e aplica-se distorções controladas:



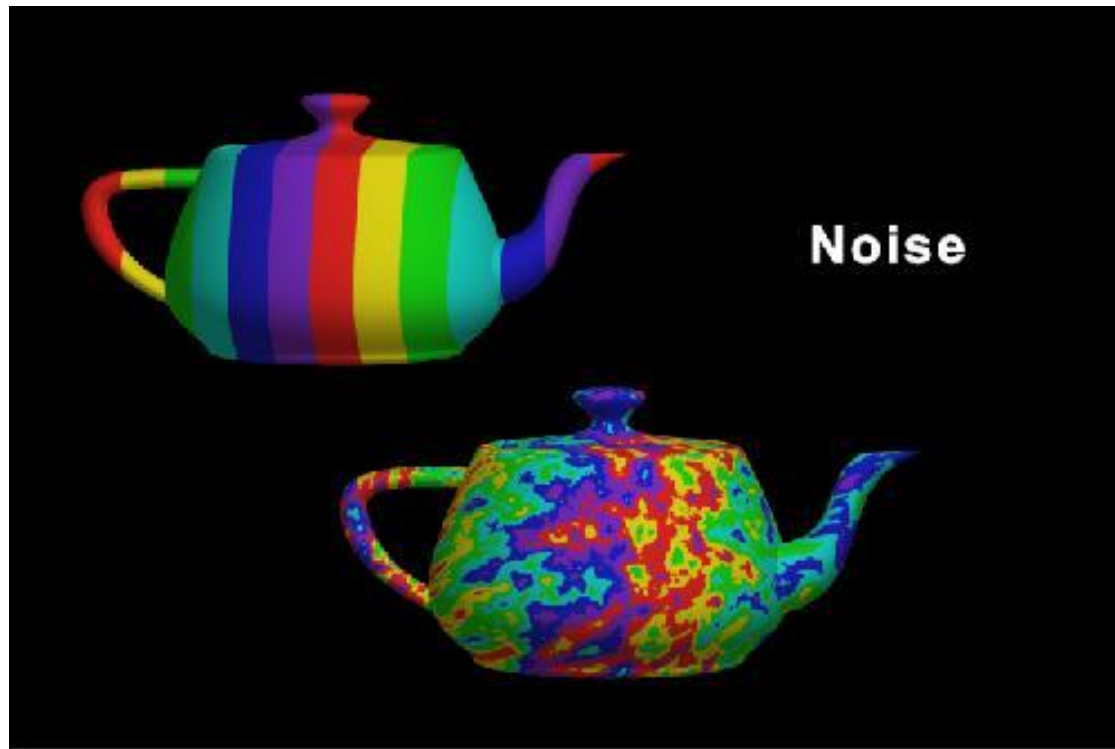
Texturização Sólida

Objetos parecem esculpidos no material.



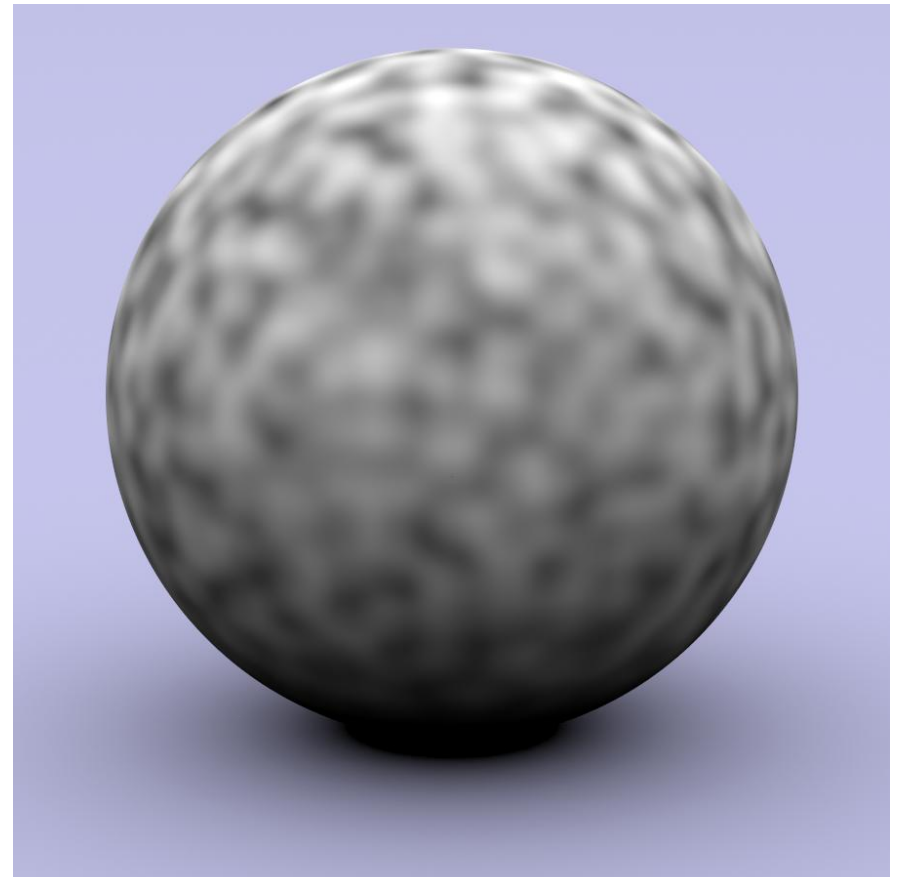
Noise

- ▶ Real-world objects have both regularity and irregularity in nature.

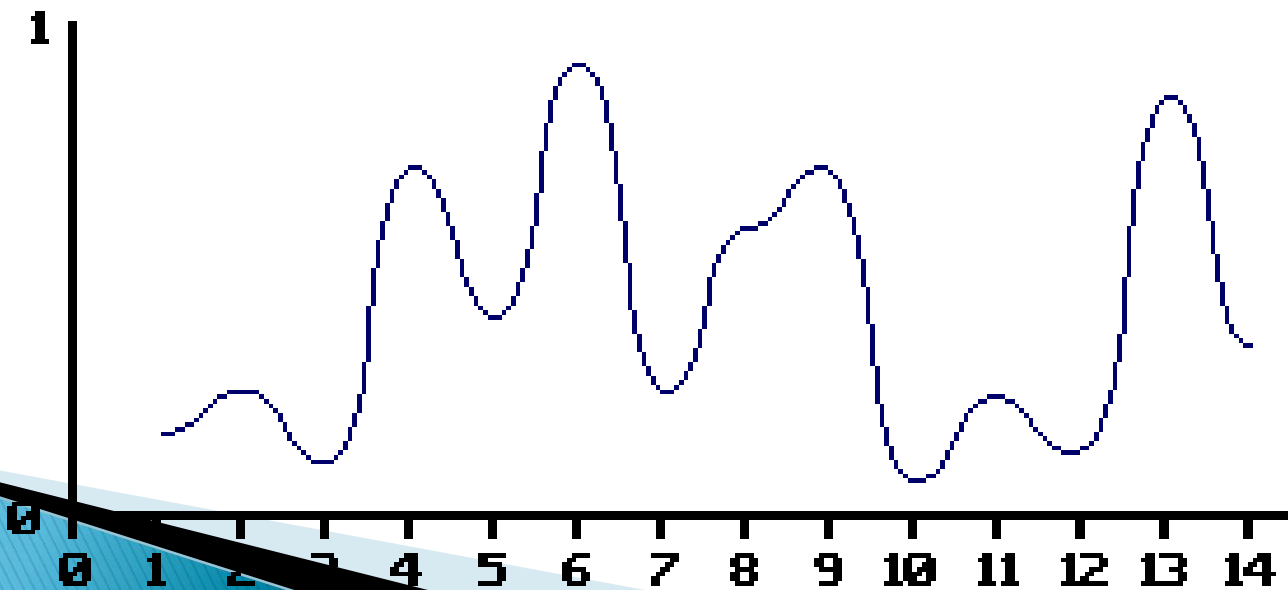
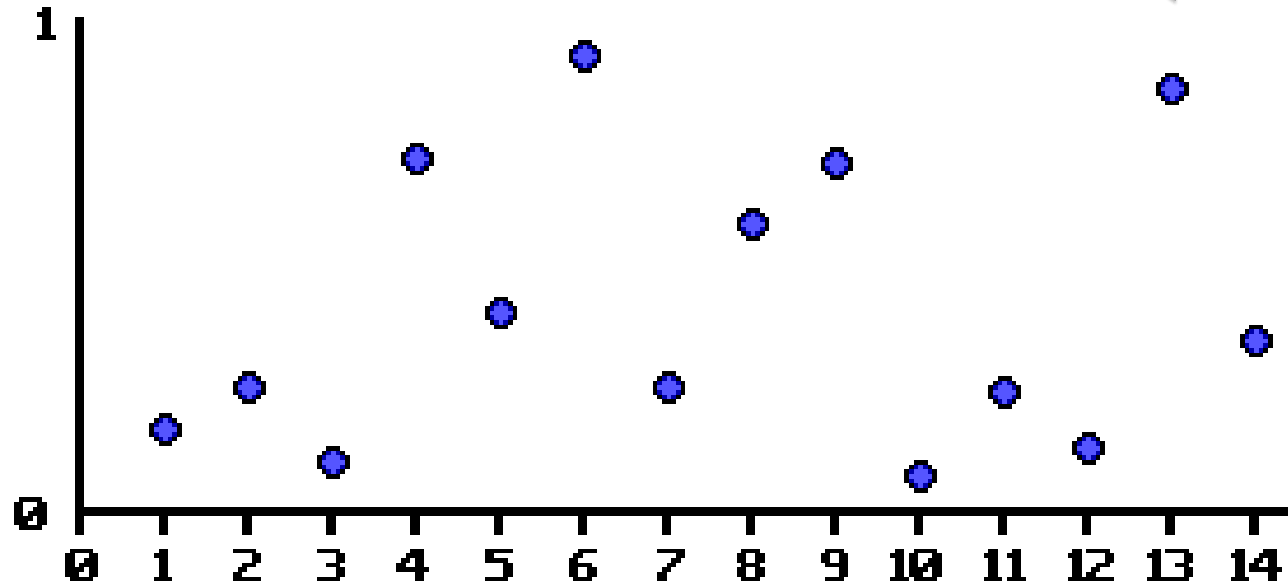


Perlin noise

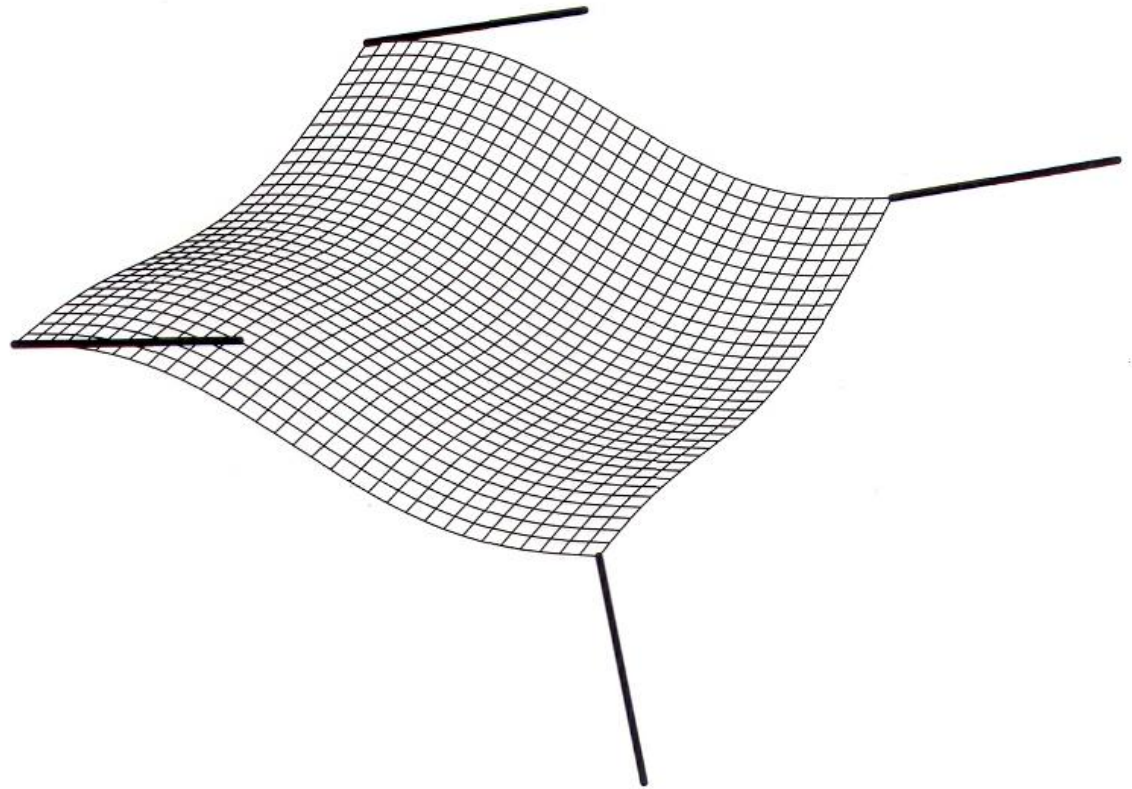
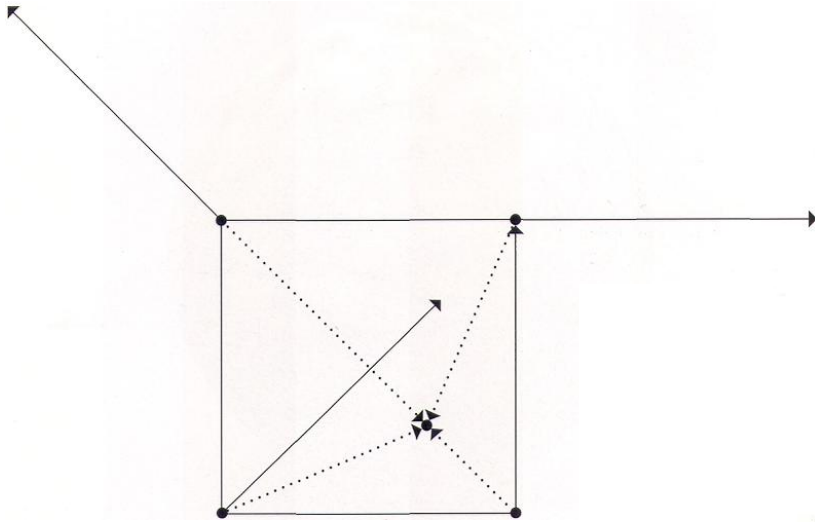
- ▶ Developed by Ken Perlin for TRON (1982)
- ▶ TRON is the first 3D shaded graphics in a Hollywood film
- ▶ Coherence randomness



Smooth out randomness (1D)

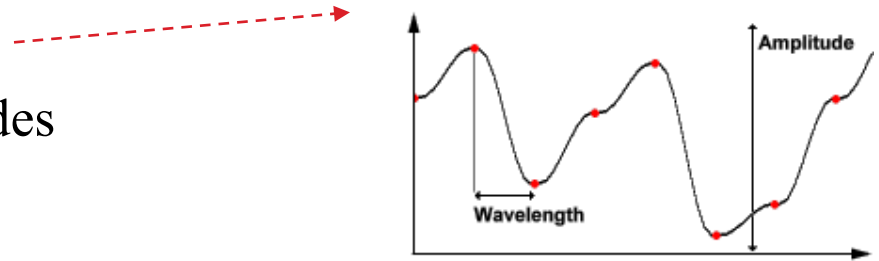


Smooth out randomness (2D)



(1-D) Perlin Noise Function :

- Get lots of such smooth functions, with various frequencies and amplitudes



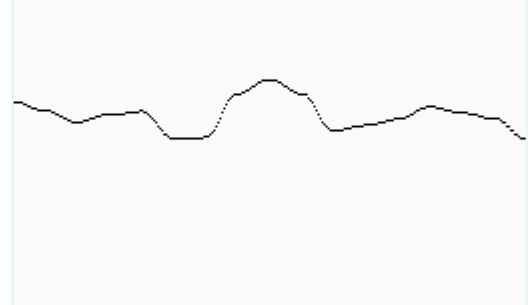
Amplitude : 128
frequency : 4



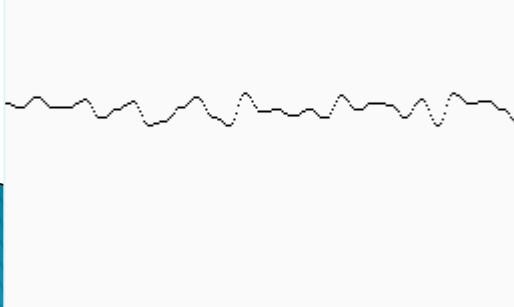
Amplitude : 64
frequency : 8



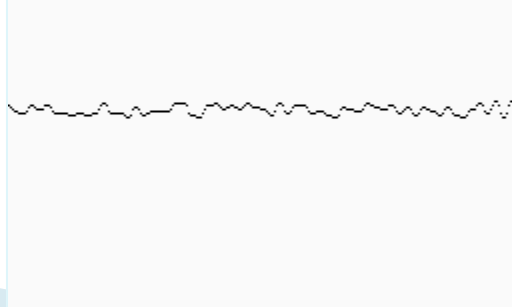
Amplitude : 32
frequency : 16



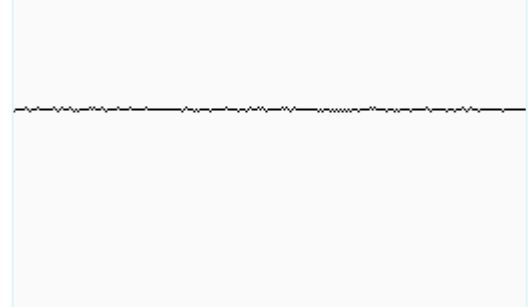
Amplitude : 16
frequency : 32



Amplitude : 8
frequency : 64

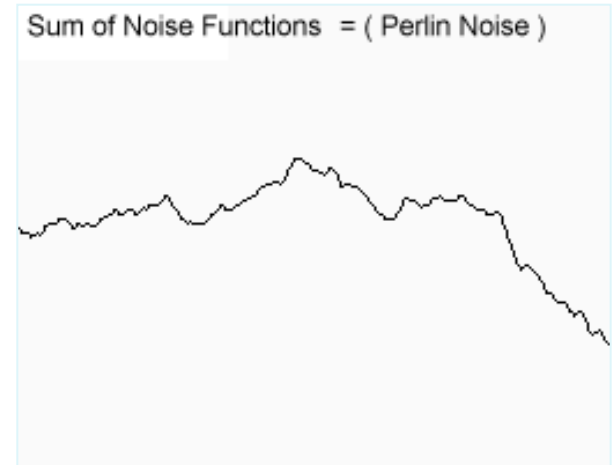
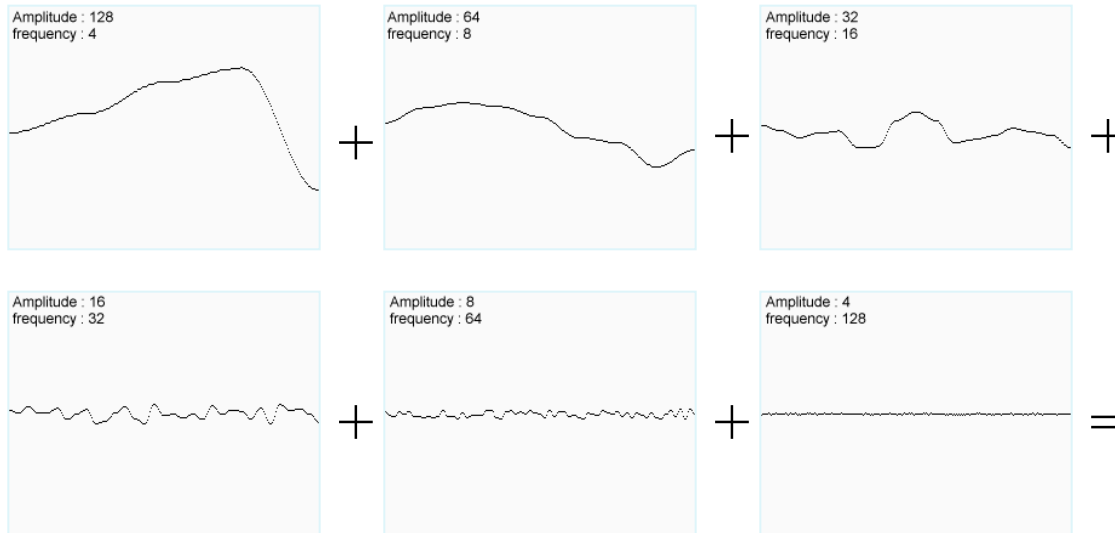
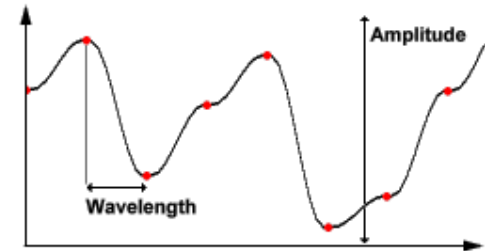


Amplitude : 4
frequency : 128

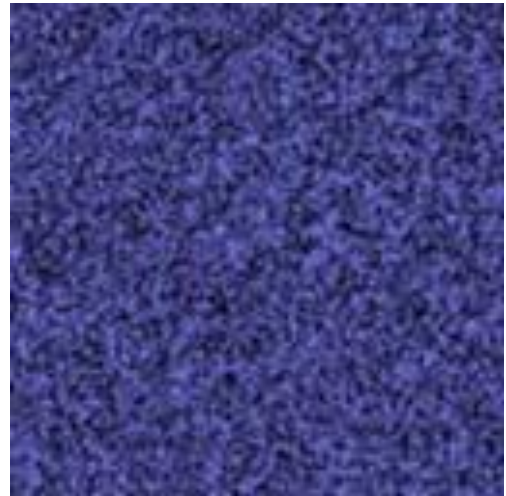
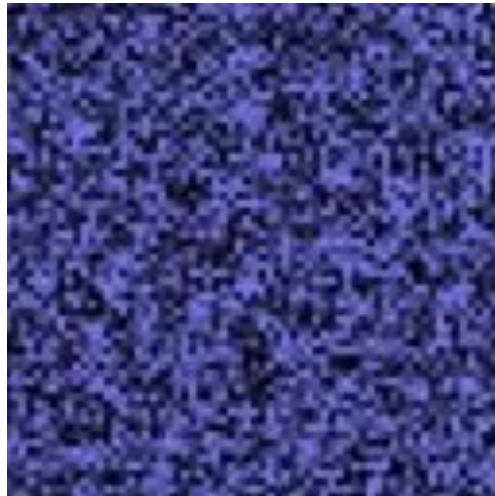
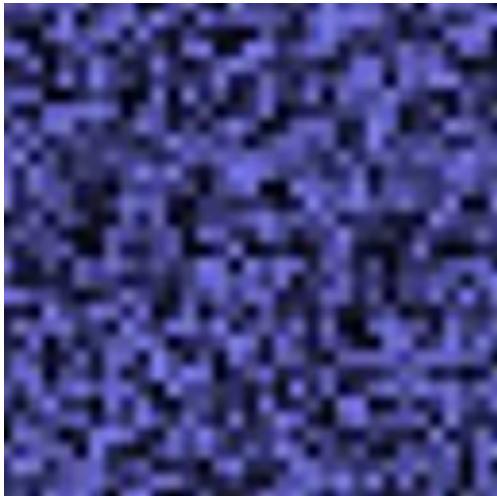
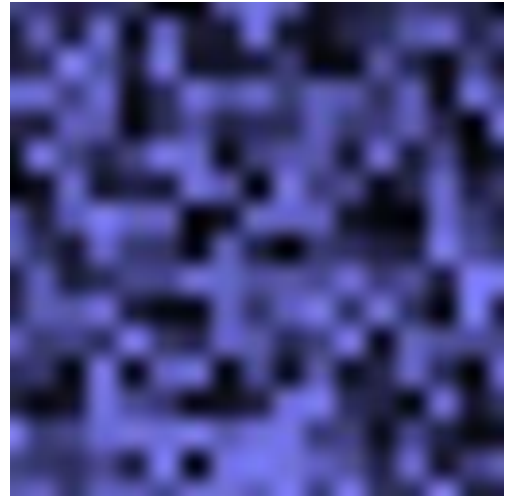
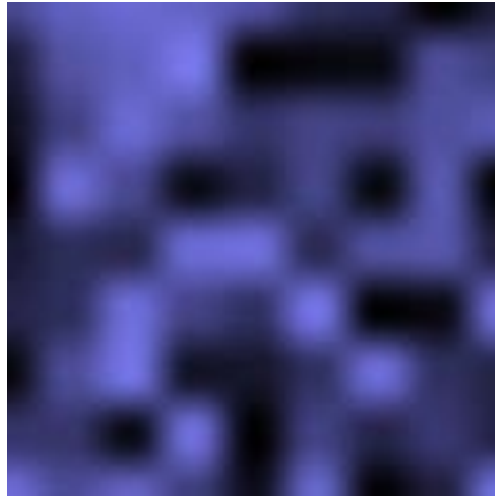
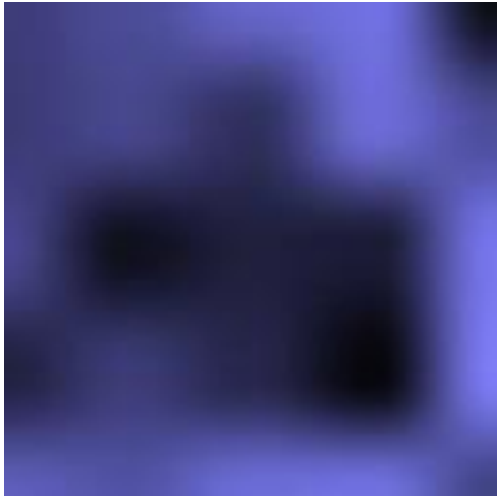


(1-D) Perlin Noise Function :

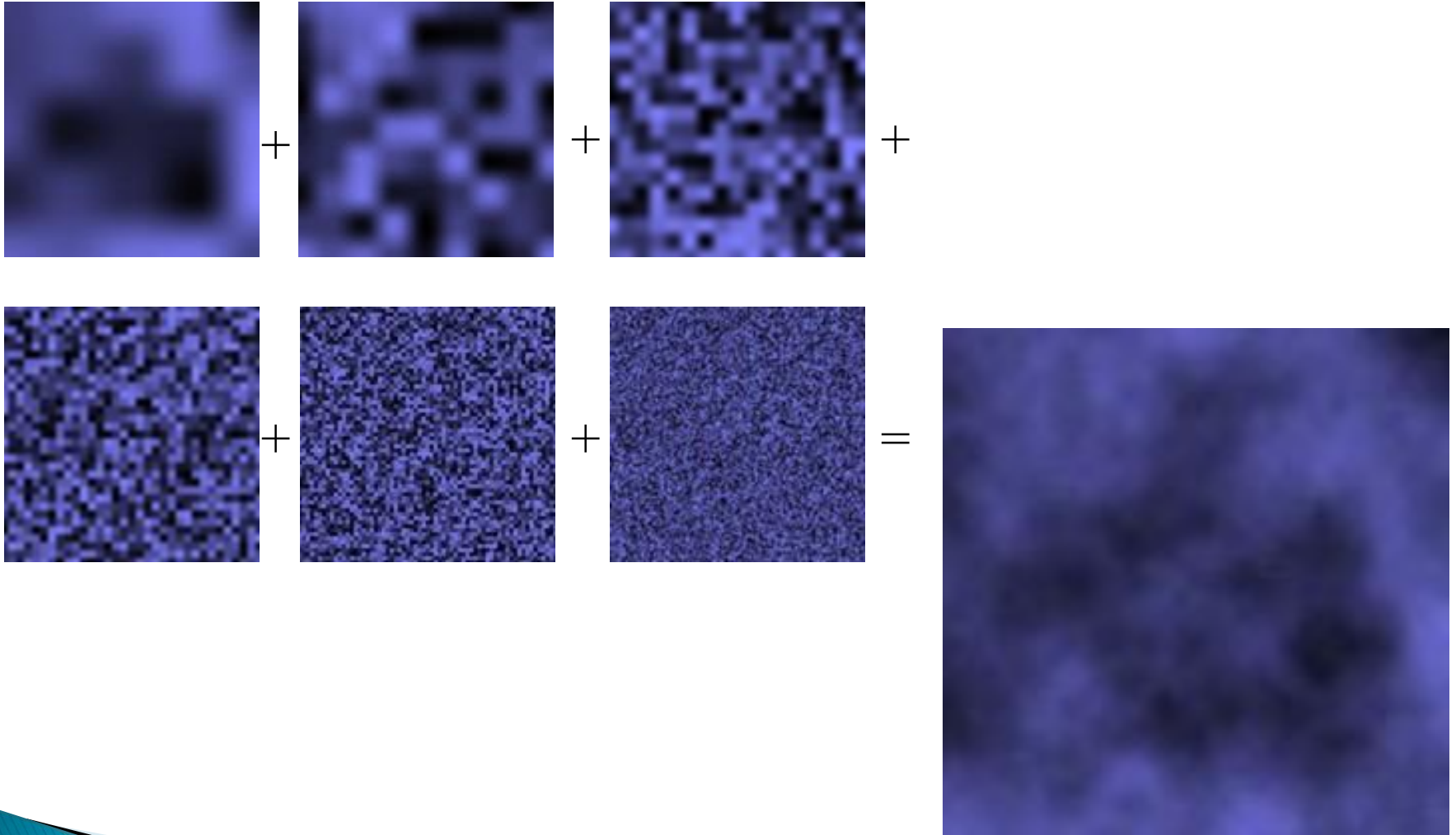
- Get lots of such smooth functions, with various frequencies and amplitudes
- Add them all together to create a nice noisy function



(2-D) Perlin Noise Function



(2-D) Perlin Noise Function



Marble

- ▶ Perturb texture coordinates before using another texture or lookup table

